A Multiscale Anisotropic Thermal Model of Chiplet Heterogeneous Integration System

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***Abstract*— Due to a variety of limitations on the system-on-chip (SoC), the microelectronics industry is now facing challenges and making slow progress in recent years. With architecture design and advanced packaging advantages, chiplet heterogeneous integration (CHI) systems have become a promising solution to long-lasting hardship. However, high power consumption in CHI systems generates massive heat and makes thermal design a demanding task. Therefore, an accurate tool for thermal simulation is indispensable in the design flow. In this article, a multiscale anisotropic thermal model is proposed for the CHI systems. It considers the feature-scale thermal conductivities of different materials to predict the package-scale steady-state temperature fields. Specifically, the local material composition and thermal conductivity of redistribution layers (RDLs) are extracted from design layout files by constructing an equivalent thermal conductivity algorithm of local feature structures. As for through silicon via (TSV) and bump arrays, the anisotropic distributions of thermal conductivity can also be derived with equivalent algorithms. Other structures are considered homogeneous blocks to significantly reduce the computational expense without losing the generality of the proposed model. Compared with the previous isotropic thermal model of CHI systems, the present multiscale anisotropic thermal model is proven to make temperature prediction and hotspot detection more reliable. With this tool, the reliability problems that are unpredictable and obscure for isotropic thermal models can be identified in advance, and more reasonable design space can be explored in the design flow of the CHI systems.**

***Index Terms*— Anisotropic, chiplet heterogeneous integration (CHI), equivalent algorithm, steady-state, thermal conductivity, thermal simulation.**

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I. Introduction

IN THE last decades, system-on-chip (SoC) has become the most popular form of microelectronic systems that pursue super high performance, such as CPUs and GPUs. As a result, the integration degree of functional modules soars, and the design complexity proliferates. On the other hand, the advanced process of manufacturing makes slow progress due to physical limitations. In addition, increasingly large areas of monolithic chips lead to yield loss, making the cost higher and higher. The aforementioned factors make it hard to continue Moore's law. The scaling-down rule of CMOS devices hits an unprecedented bottleneck.

In recent years, chiplet heterogeneous integration (CHI) has emerged. Typically, chiplets are integrated on an active or passive interposer instead of a monolithic substrate [1]. It enables heterogeneous IP reuse [2] and exhibits large prosperity in continuing Moore's law. By dividing a large chip into several chiplets, which are available from IP providers, designers can decrease the complexity of design work and the chip areas. The decrease in chip areas means that the yield loss can be significantly mitigated [3]. Furthermore, the design framework of the CHI systems does not entail state-of-the-art fabrication. Therefore, CHI is an alternative and promising solution to problems haunting the whole industry.

However, the high performance of CHI systems is at the cost of massive power consumption, even hundreds of Watts [4]. As a result, considerable heat is generated and imposes huge pressures on the heat dissipation. Thus, it makes the thermal design a demanding task. Unreliability, even failure by overheating, may happen if the thermal design does not meet the requirements [5]. Therefore, a satisfactorily accurate tool for thermal simulation is indispensable in the design flow of the CHI systems. To take thermal simulation accuracy to a new level, it is necessary to extract the thermal conductivity of the local design pattern structures of the redistribution layer (RDL), through silicon vias (TSVs) and bumps. Unfortunately, to the best of our knowledge, there are still no efficient thermal models reported that can handle the complicated design structures while considering the anisotropic distributions of thermal conductivity in the CHI systems.

In this article, a multiscale anisotropic thermal model is developed to extract the intricate distributions of the thermal conductivity of the complicated blocks (RDL, TSVs, and bumps), predict the package-scale steady-state temperature fields, and detect the hotspots of the CHI systems.

The contributions of the present thermal model are as follows.

1) With this model, the local feature-scale design patterns of RDL, TSVs, and bumps can be extracted from the design layout files to calculate the anisotropic thermal conductivity. Thus, the feature-scale physical parameters, which are more realistic and reasonable than the uniform ones, are available for the subsequent thermal simulation.
2) While RDL, TSVs, and bumps are handled with the anisotropic method, other components, such as the organic substrate, solder bumps, and the heat sink, are treated with the isotropic method. The method of selective computing anisotropy enables the multiscale anisotropic thermal model to deal with temperature prediction problems at different scales with flexibility.
3) An investigation that takes into account the design pattern effect of RDL, TSVs, and bumps proves that the proposed multiscale anisotropic thermal model can ameliorate the accuracy of temperature hotspot identification in the CHI systems.
4) Based on the ameliorated hotspot identification, the present multiscale anisotropic thermal model can avoid potential temperature-dependent unreliability, which is difficult for the previous isotropic thermal model to recognize. Meanwhile, the hotspots that are diagnosed by the isotropic model wrongly can be amended. Thus, the complexity of the thermal optimization can be mitigated.

The remainder of this article is organized as follows. Section II presents the previous work. The previous isotropic thermal model is clarified, and its problems are investigated in Section III. Section IV describes the model guidelines of the CHI systems. Section V illustrates the methodology for calculating the anisotropic thermal conductivities of the RDL, TSVs, and bumps. The proposed anisotropic thermal simulation model is verified and evaluated in Section VI. In Section VII, a case study is provided to expatiate the differences between the isotropic and anisotropic models. Finally, conclusions are given in Section VIII.

II. Related Work

Numerous researches have been proposed to build up efficient thermal models to simulate the microelectronic systems. For instance, alternating direction implicit (ADI) presents a more efficient method for transient thermal simulation compared with finite element method (FEM) [6], [7], [8]. Based on machine learning and neural networks, efficient thermal models can also be obtained [9], [10]. Meanwhile, 3-D-interlayer cooling emulator (ICE) enables the prediction of the transient temperature fields of 3-D chips in both air and liquid cooling conditions [11], [12], [13], [14]. *Therminator and Therminator*2 are fast thermal simulators for portable devices such as smart phones and tablets [15], [16]. However, these thermal models cannot be directly applied to the CHI systems.

As for the CHI systems, our previous work proposed methodology for both steady-state and transient thermal simulations using finite difference method [17], [18], [19]. Nevertheless, components with intricate design pattern structures, such as RDL, TSVs, and bumps, are oversimplified as homogeneous blocks with isotropic thermal conductivities. Those models ignore the local design features by assumptions and can incur simulation accuracy loss in both thermal conductivity and temperature prediction.

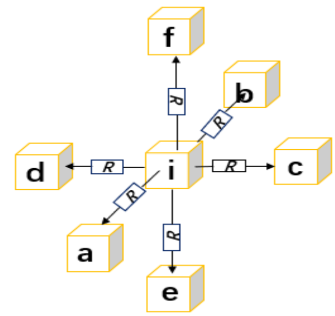

Fig. 1. Grid i and its six adjacent grids [19].

For the pursuit of anisotropy, *HotSpot* has been extended to 7.0, which can be utilized for 2.5-D CHI thermal simulations using 3-D grid mode and heterogeneous resistivity option, whereas the anisotropic resistivity is limited to block level and the organic substrate and solder bumps have to be regarded as adiabatic surfaces [20], [21], [22]. *Overheat*, developed by Mentor, one of the coupled simulators for electro-thermal simulation, can extract design pattern structures from layout files [23]. However, this model is limited to monolithic chips and takes into account just one heat transfer path where the heat sink is simplified as a cuboid. Therefore, both tools cannot be directly applied to the CHI systems with consideration of the local design structures of bumps, TSVs, and RDL.

For the above problems and limitations, a new multiscale anisotropic thermal model is developed in this article to enable the package-scale steady-state thermal simulation of the CHI systems.

III. Previous Isotropic Thermal Model and Its Problems

The previous thermal model in [19] is the prerequisite of the present multiscale anisotropic thermal model. In order to clarify the model, its detailed framework is given as follows.

First, it can be assumed that there are two thermal grids (i and j). Based on the well-known thermal-electrical duality, Ohm's law can be transformed into its thermal form as the following equation:

$$g_{i,j}\left(T_i - T_j\right) = Q_{i,j} \tag{1}$$

where $g_{i,j}$ represents the thermal conductance (counterpart of electrical conductance) between grids i and j, $T_{i(j)}$ refers to the steady-state temperature (counterpart of electrical potential) of grid i (j), and $Q_{i(j)}$ is the rate of heat flow (counterpart of electrical current) from grid i to j [19].

Furthermore, we can assume that grid i has six adjacent grids named a–g (R represents the thermal resistance between two adjacent grids), as shown in Fig. 1 [19]. The left-hand side represents the rate of the heat flow from grid i to j

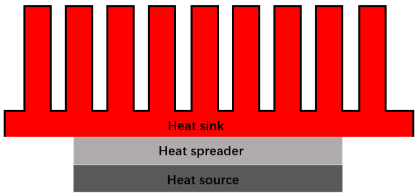

Fig. 2. Schematic architecture utilized to investigate the simulation problems of isotropic thermal models.

(counterpart of the electrical current flowing out a circuit node) and P_i refers to the power consumption of grid i (counterpart of the electrical current flowing into a circuit node). Thus, the Kirchhoff's current law can be transformed into its thermal form as well [19]

$$\sum_{j=a}^{f} g_{i,j}\left(T_i - T_j\right) = P_i. \tag{2}$$

Equation (2) is the steady-state heat equation for an interior grid where heat is transported in the form of conduction. For a boundary grid where heat is transferred in the forms of conduction and convection, we can derive its steady-state heat equation by replacing $g_{i,j}$ and T_j with $g_{i,\mathrm{amb}}$ and T_{amb}, respectively. T_{amb} is the temperature of the ambient air. $g_{i,\mathrm{amb}}$ is formulated as the following equation:

$$g_{i,\mathrm{amb}} = g_i // hA \tag{3}$$

where g_i refers to the thermal conductance of grid i, h represents the heat transfer coefficient, and A symbols the area exposed to the ambient air [19].

In a CHI system, all the grids form an equivalent thermal resistance network. By writing the steady-state heat equations of all the grids ($1 \sim n$) and organizing them in a matrix form, the system of linear equations can be written as the following equation:

$$\boldsymbol{M}_{\mathbf{conductance},n\times n} \cdot \boldsymbol{V}_{\mathbf{temp},n\times 1} = \boldsymbol{V}_{\mathbf{power},n\times 1} \tag{4}$$

where $\boldsymbol{M}_{\mathbf{conductance},n\times n}$ refers to the thermal conductance matrix of CHI systems. $\boldsymbol{V}_{\mathbf{temp},n\times 1}$ and $\boldsymbol{V}_{\mathbf{power},n\times 1}$ represent the temperature and power vectors of CHI systems, respectively [19]. The steady-state temperature fields of CHI systems can be acquired by solving (4). Based on these principles and deduction, the steady-state isotropic thermal model for CHI systems is constructed and validated against FEM in [19]. Its simulation errors are less than 1.8%.

However, in such isotropic thermal models [17], [18], [19], [24], the equivalent thermal conductivity is always calculated as an average according to the area ratio, which can incur accuracy loss in the thermal simulation. To investigate the simulation problems of such isotropic thermal models, a simple and hypothetical structure is illustrated in Fig. 2. From the bottom to up, there is a heat source, a heat spreader, and a heat sink. The material of the heat source is silicon ($k_{\mathrm{normal}} = k_{\mathrm{planar}} = 130\ W/(mK)$) and that of the heat sink is aluminum ($k_{\mathrm{normal}} = k_{\mathrm{planar}} = 237\ W/(mK)$). As for the heat spreader, it includes silicon dioxide ($k_{\mathrm{normal}} = k_{\mathrm{planar}} = 1.5\ W/(mK)$) as the substrate. In addition, 20 parallel copper wires ($k_{\mathrm{normal}} = k_{\mathrm{planar}} = 401\ W/(mK)$) are also included in the heat spreader. The heat source and heat spreader are set to represent a 2-D die. The aforementioned thermal conductivities are the intrinsic physical parameters of materials and have been widely used in thermal simulations [4], [5], [17], [18], [19], [20], [21], [22].

TABLE I
DIMENSIONS OF COMPONENTS

NAME	DIMENSIONS (mm)
Heat source	11.7 × 11.7 × 0.05
Heat spreader	11.7 × 11.7 × 0.1
Metal wires	9.7 × 0.2 × 0.1
Base	60 × 60 × 1
Fins	2 × 60 × 6, space = 2

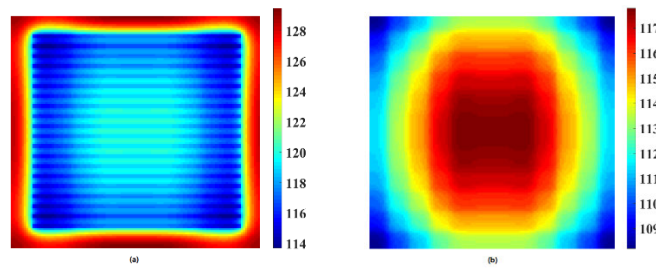

Fig. 3. (a) Actual and (b) isotropic thermal maps of the top surface of the heat source.

The dimensions of the three components are shown in Table I. The power consumption of the heat source is set to 50 W. The diabatic surfaces exposed to the ambient air (25 °C) are highlighted in black lines. The heat transfer coefficient h is equal to 50 $W/(m^2K)$. These simulation parameters are set according to [19].

The structure in Fig. 2 is thermally simulated with the thermal model which was developed in [19]. In this thermal simulation, the thermal conductivities of the heat source and heat sink are isotropic. As for the anisotropic heat spreader, no equivalent algorithm or model simplification is adopted to obtain its thermal conductivity. On the contrary, the actual thermal conductivity of the heat spreader is considered in the full-structure thermal simulation. The thermal map of the heat source's top surface is shown in Fig. 3(a). It serves as the ground-truth temperature field and is named as the actual thermal map.

In contrast, the equivalent thermal conductivity of the heat spreader can be calculated as an average according to the area ratio in the z-direction, which can be expressed as the following equation:

$$k_{\text{heat spreader}} = \frac{S_{\text{metal}}}{S_{\text{heat spreader}}} \times k_{\text{metal}} + \frac{S_{\text{dioxide}}}{S_{\text{heat spreader}}} \times k_{\text{dioxide}}. \tag{5}$$

In this way, the equivalent thermal conductivity of the heat spreader can be calculated as 114.7 $W/(mK)$. With this isotropic equivalent thermal conductivity, the thermal map of

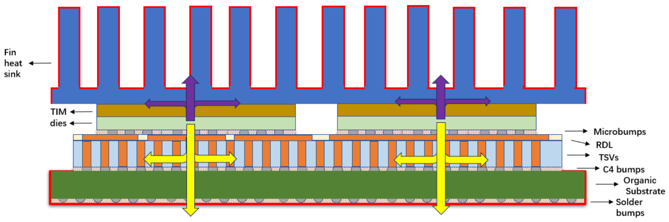

Fig. 4. Architecture of CHI systems and the two heat transfer paths.

the heat source's top surface is called the isotropic thermal map and is illustrated in Fig. 3(b).

In Fig. 3(a) and (b), the temperature values and locations of hotspots are obviously different, which implies that the isotropic thermal model can lead to considerable accuracy loss. Potential hotspots may survive and thus result in unexpected thermally related unreliability. These problems originated from the assumption that complicated components can be regarded as homogeneous blocks. In order to make the thermal simulation more instructive and accurate, an anisotropic thermal model is indispensable to capture the local thermal conductivities of design pattern structures.

IV. Model Guidelines

The representative architecture of CHI systems is shown in Fig. 4. It is assumed that electricity is only consumed in dies and ultimately transformed into heat. In nondie components, the self-heating effect is neglected for model simplicity. Therefore, only dies serve as heat sources in the CHI systems. Heat is evenly generated in each die. Subsequently, the generated heat is removed upward and downward into the ambient air through two transfer paths (path A in purple and path B in yellow), as shown in Fig. 4. Along path A, heat flows upward through dies, thermal interface material (TIM), heat sink and finally into the ambient air. Along path B, heat flows downward through dies, microbumps, RDL, TSVs, C4 bumps, organic substrate, solder bumps, and finally into the ambient air.

In addition, RDL, TSVs, and bumps have countless wires, cylinders, or spheres [25]. The position distributions of these design pattern structures affect their local thermal conductivities and the simulation results. Hence, it is critical to extract specific structure features of the RDL, TSVs, and bumps. Fortunately, layout files contain the geometry features of these materials, which can be utilized to obtain the local thermal conductivity distributions. Therefore, components are classified into two types: isotropic and anisotropic components. The isotropic components refer to the components whose thermal conductivity can be regarded as isotropic, such as the heat sink, TIM, organic substrate, and solder bumps. Users need to define each isotropic component's geometry parameters (3-D space coordinates, 3-D dimensions, and the number of grid cells) and thermal parameters (normal and planar thermal conductivity and power consumption). The anisotropic components refer to the components which are intricately composed and have anisotropic thermal conductivity distributions, such as microbumps, RDL, TSVs, and C4 bumps. In addition to their geometry and thermal parameters, users are supposed to offer the following extra parameters for anisotropy calculation: graphic design system II (GDSII) files of RDL and dies, design geometry parameters of bump arrays (radius, height, pitch, and size), and design geometry parameters of TSV arrays (radius of metal cylinders, thickness of insulating layer, height, pitch, and size).

At the boundaries of CHI systems, heat is assumed to be removed in the form of air convection. The heat radiation effect is neglected for model simplicity. Therefore, users are supposed to define the systems' diabatic surfaces, which are exposed to the ambient air (such as the highlighted surfaces of the heat sink, organic substrate, and solder bumps in Fig. 4). As for the ambient air, it is assumed that its temperature is isotropic, constant and not affected by CHI systems. In both Sections III and VII, the temperature of the ambient air is set to 25 °C.

V. Anisotropic Methodology

In this section, the anisotropic methodology is proposed. Specifically, a method is presented in Section V-A to extract boundaries and paths from GDSII files. Based on Section V-A, Section V-B illustrates the way in which the material composition of each grid is calculated. Equivalent algorithms are developed in Sections V-C and V-D to calculate the local equivalent thermal conductivities according to the local material compositions. Finally, the derived anisotropic distributions of thermal conductivity are introduced to the construction of the thermal conductance matrix, $\boldsymbol{M}_{\mathbf{cond},n\times n}$ in (4), to achieve anisotropic thermal simulations of CHI systems.

A. Extracting Boundaries and Paths From GDSII Files

For the fabrication of integrated circuits, a layout file is an essential prerequisite for all the masks and reticles since it contains the geometrical data of every single layer. Therefore, information about the design pattern structures can be extracted from layout files and interpreted for the calculation of the local material compositions.

Binary GDSII files are one of the mainstream formats in the microelectronics industry. There are two fundamental geometrical elements called *boundary* and *path* in GDSII files. A boundary is a polygon defined by all of its vertices, while a path is a finitely wide line presented by its central points and width. Data about the two elements are presented in a format called a *record*. Each record has a 4-byte header which declares its size and function. In addition, there are optional contents following each header to carry specific information (layer number, coordinates of vertices, width, and so on). Therefore, information about polygons and paths in every single layer can be extracted from GDSII layout files by identifying and interpreting those records.

After they are identified and interpreted, specific structure features, such as the layer in which wires and vias are located, size, and planar coordinate, are finally available for subsequent calculation of the local material compositions.

B. Calculating Local Material Compositions

In Fig. 5, polygons in orange are geometrical elements extracted from GDSII files, such as metal wires and vias in

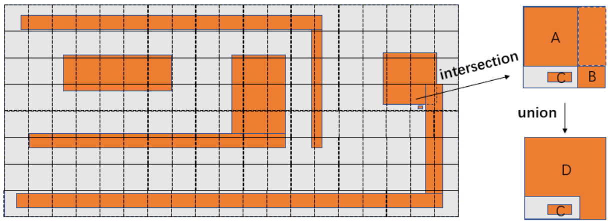

Fig. 5. Geometrical elements and material composition of each grid.

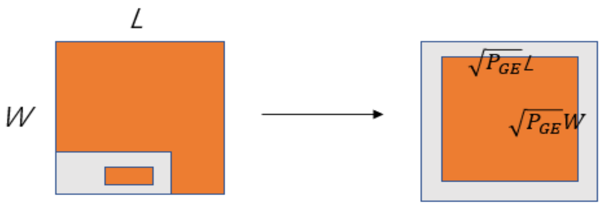

Fig. 6. Area transformation in a grid.

RDL. The silicon dioxide is presented in gray as the substrate. Based on the extracted geometrical elements, the material composition of each grid (in dash lines) can be derived. First, for a certain grid, intersection operations are executed in it and all of the geometrical elements to obtain shapes enclosed in the grid, like rectangles A, B, and C in the magnified gird of Fig. 5. Subsequently, union operations are performed between the enclosed shapes in this grid to avoid redundant calculation of overlapped areas. For instance, rectangles A and B have a region in common, which is outlined in the dashed line. Union operation between rectangles A and B forms hexagon D in Fig. 5. In general, union operation continues until there is no overlapped region between any pair of enclosed shapes in a grid. Eventually, areas of all the separate shapes are measured and added up to calculate the percentage of different materials. For the grid in Fig. 5, the step is expressed by the following equations:

$$P_{\mathrm{GE}} = \frac{S_{\mathrm{C}} + S_{\mathrm{D}}}{S_{\mathrm{G}}} \times 100\% \tag{6}$$

$$P_{\mathrm{S}} = \frac{S_{\mathrm{G}} - S_{\mathrm{C}} - S_{\mathrm{D}}}{S_{\mathrm{G}}} \times 100\%. \tag{7}$$

where S_{C}, S_{D} and S_{G} refer to the areas of rectangle C, hexagon D, and the grid, respectively. P_{GE} and P_{S} represent the material percentages of geometrical elements and the substrate. After all the grids are handled in this way, the local material compositions of the whole GDSII layer can be figured out. According to the derived local material compositions, the anisotropic thermal conductivity of the graphic design system (GDS) layer can be obtained with the methodology proposed in Section V-C.

C. Local Equivalent Thermal Conductivity of RDL

As depicted in Fig. 6, for a grid with material percentage P_{GE}, the geometrical elements can be transformed into a single and concentric rectangle with the same aspect ratio. The deduction of the equivalent algorithm for local equivalent thermal conductivity is based on this transformation.

As for the normal direction, heat can flow through the central rectangle in orange or the peripheral area in gray,

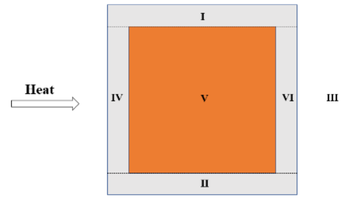

Fig. 7. Different regions of the heat transfer path in x-direction.

which means that the two transfer paths are parallel. Therefore, the normal thermal conductance in the grid, G_{normal} can be expressed as the following equation:

$$\begin{aligned} G_{\mathrm{normal}} &= \frac{k_{\mathrm{GE,normal}} \times P_{\mathrm{GE}} \times S_{\mathrm{G}}}{h_{\mathrm{G}}} + \frac{k_{\mathrm{S,normal}} \times P_{\mathrm{S}} \times S_{\mathrm{G}}}{h_{\mathrm{G}}} \\ &= \frac{k_{\mathrm{eq,normal}} \times S_{\mathrm{G}}}{h_{\mathrm{G}}} \end{aligned} \tag{8}$$

where $k_{\mathrm{GE,normal}}$, $k_{\mathrm{S,normal}}$, and h_{G} refer to the normal thermal conductivities of geometrical elements, substrate, and the height of the grid, respectively. Thus, the equivalent normal thermal conductivity of the grid, $k_{\mathrm{eq,normal}}$ can be written as the following equation:

$$k_{\mathrm{eq,normal}} = P_{\mathrm{GE}} \times k_{\mathrm{GE,normal}} + P_{\mathrm{S}} \times k_{\mathrm{S,normal}}. \tag{9}$$

Obviously, it is a linearly weighted sum of $k_{\mathrm{GE,normal}}$ and $k_{\mathrm{S,normal}}$.

Taking the heat transfer in the x-direction into account, we can divide a grid into three regions named I, II, and III, as shown in Fig. 7. Furthermore, regions III can be partitioned into subregions IV, V, and VI. Apparently, regions I, II, and III are parallel as heat flows through these regions, while subregions IV, V, and VI are in series.

Therefore, the thermal conductance in the grid in the x-direction, G_x, can be written as the following equation:

$$\begin{aligned} G_x = &\frac{\frac{1}{2} \times \left(W - \sqrt{P_{\mathrm{GE}}}W\right) \times h_{\mathrm{G}} \times k_{\mathrm{S,planar}}}{L} \\ &+ \frac{\frac{1}{2} \times \left(W - \sqrt{P_{\mathrm{GE}}}W\right) \times h_{\mathrm{G}} \times k_{\mathrm{S,planar}}}{L} \\ &+ G_{\mathrm{III},x} = \frac{k_{\mathrm{eq},x} \times W \times h_{\mathrm{G}}}{L} \end{aligned} \tag{10}$$

where $k_{\mathrm{S,planar}}$ refers to the planar thermal conductivity of the substrate. Meanwhile, the thermal conductance in the x-direction of region III, $G_{\mathrm{III},x}$, can be expressed as the following equation:

$$\begin{aligned} G_{\mathrm{III},x} = &\left(\frac{\sqrt{P_{\mathrm{GE}}}W \times h_{\mathrm{G}} \times k_{\mathrm{S,planar}}}{\frac{1}{2} \times \left(L - \sqrt{P_{\mathrm{GE}}}L\right)}\right) // \\ &\left(\frac{\sqrt{P_{\mathrm{GE}}}W \times h_{\mathrm{G}} \times k_{\mathrm{S,planar}}}{\frac{1}{2} \times \left(L - \sqrt{P_{\mathrm{GE}}}L\right)}\right) // \\ &\left(\frac{\sqrt{P_{\mathrm{GE}}}W \times h_{\mathrm{G}} \times k_{\mathrm{GE,planar}}}{\sqrt{P_{\mathrm{GE}}}L}\right) \end{aligned} \tag{11}$$

where $k_{\mathrm{GE,planar}}$ represents the planar thermal conductivity of the geometrical elements. Based on (10) and (11), the

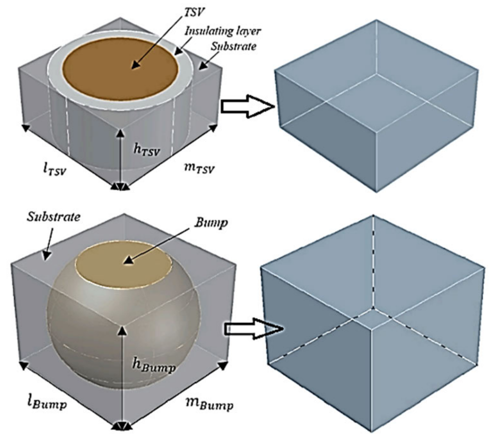

Fig. 8. Thermally equivalent transformation of a single TSV and bump [17].

equivalent thermal conductivity in the grid in the x-direction, $k_{\text{eq},x}$, can be further presented as the following equation:

$$k_{\text{eq},x} = \left(1 - \sqrt{P}_{\text{GE}}\right) \times k_{\text{S,planar}} + \frac{\sqrt{P}_{\text{GE}} \times k_{\text{GE,planar}} \times k_{\text{S,planar}}}{\left(1 - \sqrt{P}_{\text{GE}}\right) \times k_{\text{GE,planar}} + \sqrt{P}_{\text{GE}} \times k_{\text{S,planar}}}. \tag{12}$$

Its counterpart in y-direction, $k_{\text{eq},y}$, which is equal to $k_{\text{eq},x}$, can be obtained in the same way. In summary, the equivalent planar thermal conductivity in the gird can be represented as the following equation:

$$k_{\text{eq,planar}} = k_{\text{eq},x} = k_{\text{eq},y}. \tag{13}$$

As for both wire and via layers in RDL, the anisotropic distributions of thermal conductivity can be figured out with (9), (12), and (13).

D. Local Equivalent Thermal Conductivity of the TSV and Bump Layers

It should be noted that the anisotropic methodology described in Sections V-B and V-C applies to polygons such as metal wires and vias in RDL. It cannot directly handle circular TSVs and bumps. To calculate its anisotropic material positions and thermal conductivity, every single TSV or bump needs to be transformed into a desired polygon with high accuracy. With the equivalent thermal conductivity algorithm, which is proposed and validated in [17], every single TSV or bump can be transformed into its thermally equivalent cuboid according to its geometries and thermal parameters, such as the radius of a single TSV or bump, the thickness of the insulating layer and thermal conductivities of different materials (copper, insulator, substrate, and underfill). As illustrated in Fig. 8, in this way, a single TSV or bump can be replaced with its thermally equivalent cuboid with simulation errors <2% [17].

As a result, an array of TSVs or bumps can be transformed into an array of cuboids, as shown in Fig. 9. An array of cuboids makes it possible to calculate the anisotropy through the methodology described in Sections V-C. This methodology is applied to TSV or bump arrays with different design parameters (radius of TSVs or bumps, height, thickness of the insulating layer, pitch, and size).

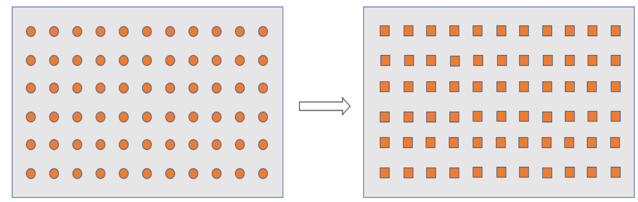

Fig. 9. TSV or bump array is transformed into an array of polygons.

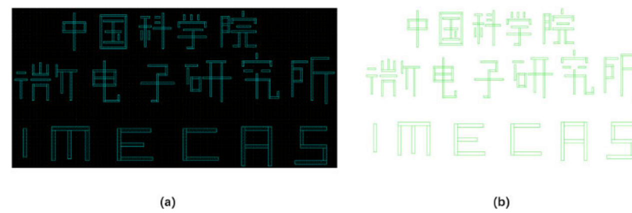

Fig. 10. (a) GDSII layout for verification. (b) Extracted patterns.

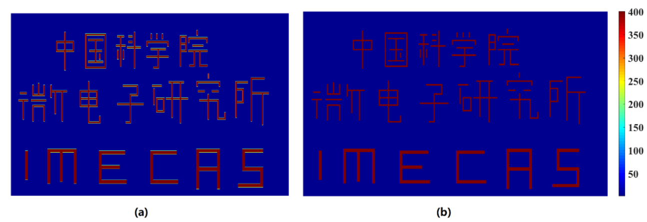

Fig. 11. Distributions of (a) normal and (b) planar equivalent thermal conductivities.

VI. Model Verification

In this section, the anisotropic methodology proposed in Section V is validated and evaluated in Section VI-A. Then, the anisotropic thermal model is evaluated with two comparative analyses in Section VI-B.

A. Methodology Validation and Evaluation

As shown in Fig. 10(a), a simple GDSII layout file is taken as a case to verify the proposed anisotropic methodologies. After the layout patterns are extracted and interpreted, all the vertices are connected to visualize all the polygons, as shown in Fig. 10(b). It can be clearly seen that the extracted patterns are identical to the original layout patterns.

Based on the extracted patterns, the anisotropic distributions of normal and planar thermal conductivity are obtained with the proposed equivalent thermal conductivity algorithms. The thermal conductivities of geometrical elements k_{GE}, and substrate k_{S} are set to 401 $W/(mK)$ and 1.5 $W/(mK)$, respectively. The results are displayed in Fig. 11. It is obvious that the distributions are highly consistent with the original GDSII layout, which verifies the equivalent thermal conductivity algorithm.

In contrast, the normal and planar conductivities are 2.039725 $W/(mK)$ and 1.502096 $W/(mK)$, if the grid number is set to 1 × 1, which corresponds to isotropic methods. Apparently, the anisotropic thermal conductivities are much

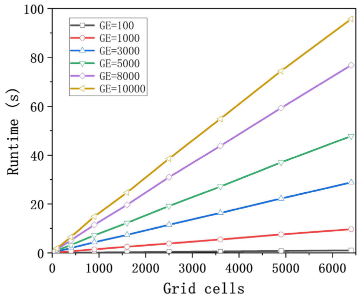

Fig. 12. Runtimes of anisotropy calculation as functions of grid cells under different geometrical elements.

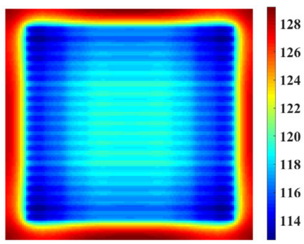

Fig. 13. Thermal map derived by the anisotropic thermal model.

closer to reality than average values which are derived with isotropic methods.

However, it takes time to obtain accurate anisotropic distributions of thermal conductivity. Therefore, to further evaluate the proposed anisotropic methodology, the runtime of the anisotropy calculation is investigated using a computer with Intel[1] Core i5-6200U CPU (2.3 GHz) and 8 GB RAM. The runtimes are plotted in Fig. 12 as functions of grid cells (granularity of thermal simulations) under different geometrical elements (polygons in a single layer of GDSII layout). It can be clearly seen that the runtimes increase as the number of both grid cells and geometrical elements go up, which suggests that overfine granularity in thermal simulation or overcomplicated circuit design in GDSII layout can lead to overlong runtime.

B. Evaluation of the Proposed Anisotropic Thermal Model

The anisotropic thermal simulation can be achieved by introducing the anisotropic thermal conductivity into the construction of the thermal conductance matrix $\boldsymbol{M}_{\mathbf{cond},n\times n}$ in (4). Specifically, the local thermal conductivity in each grid can be utilized to calculate the local thermal conductance. It means that each grid has its unique thermal conductance rather than a uniform one. Therefore, every element in the thermal conductance matrix is more accurate and realistic than that derived by isotropic thermal models.

With the structure depicted in Fig. 2, an anisotropic thermal simulation is performed to evaluate the present anisotropic thermal model. The thermal map of the heat source's top surface is displayed in Fig. 13. It can be seen that the temperature hotspots are located in the peripheral areas, which is in great accordance with the actual thermal map displayed in Fig. 3(a). In addition, the junction temperatures in Figs. 3(a), 13, and 3(b) is 129.5 °C, 129.2 °C, and 117.8 °C,

[1]Registered trademark.

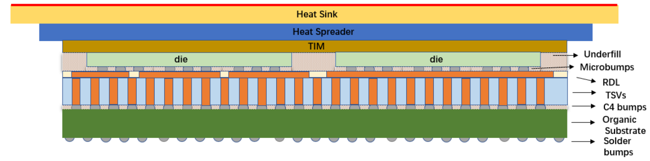

Fig. 14. CHI architecture used for the comparative analysis.

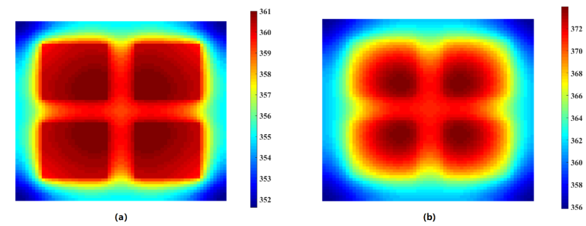

Fig. 15. Thermal maps of the RDL by (a) proposed thermal model and (b) HotSpot.

respectively. The lowest temperatures are 113.7 °C, 112.6 °C, and 108.2 °C. The comparison of the three thermal maps indicates that the present anisotropic thermal model can predict temperature fields better than the isotropic thermal model.

Due to self-consistency, the anisotropic thermal model can be extended to larger and more complicated CHI systems to obtain package-scale temperature fields. For further evaluation, a comparative analysis between the proposed thermal model and HotSpot is presented. The CHI architecture used for the comparative analysis is illustrated in Fig. 14.

Compared with Section VII, the sides of solder bumps and organic substrate are decreased to 30 mm. The heat spreader edge size is 2× larger than the interposer. The heat sink edge size is also 2× larger than the heat spreader. In addition, epoxy resin is underfilled between dies and microbumps. Furthermore, the thermal conductivities of components are the same as the normal conductivities listed in Section VII. As for the boundary conditions, the top surface of the heat sink is regarded as the only diabatic surface, as highlighted in red. Its heat transfer coefficient is set to 70 $W/(m^2K)$. The grid cells of the stacked layers are set to 64 × 64. Compared with the CHI architecture and boundary conditions described in Section VII, these modifications are made due to the following reasons: 1) in the 3-D grid mode of HotSpot, the dimensions of stacked layers must be consistent; 2) in the 3-D grid mode of HotSpot, the organic substrate and solder bumps cannot be set as diabatic surfaces; and 3) in HotSpot, there are no options to separately set the normal and planar conductivities of a block or a grid. Then, the thermal maps of the RDL are displayed in Fig. 15 (temperature unit in Kelvin). It can be clearly seen that the temperature profiles are in qualitative consistency: the regions beneath the four chiplets (heat sources) are at the highest temperatures, and temperature decreases as it goes peripherally. However, the temperature maxima and minima are different. It may be explained as follows: in HotSpot, there are four trapezoidal grids in the periphery of the heat spreader and eight trapezoidal grids in the periphery of the heat sink, as shown in Fig. 16 [5], [26]. If the heat spreader and the heat sink are too large,

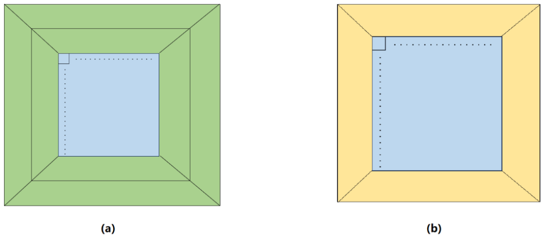

Fig. 16. Grid cells of the (a) heat sink and (b) heat spreader in HotSpot.

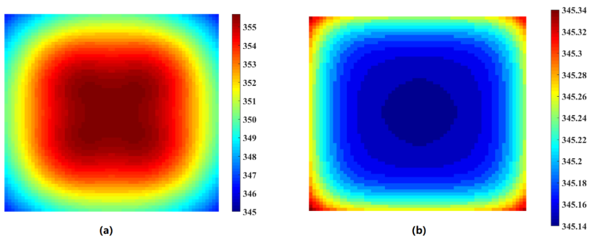

Fig. 17. Thermal maps of the central regions of the heat sink by (a) proposed thermal model and (b) HotSpot.

like this CHI architecture, the trapezoidal grids will be too coarse, which could result in accuracy loss. In addition to the thermal maps of the RDL, the thermal maps of the heat sink are also displayed in Fig. 17 to show this problem. As shown in Fig. 17(b), the temperature variation is so slight that the heat sink can be almost regarded as isothermal. As pointed out by the developers of HotSpot, the heat sink should not be fully isothermal in real cases [26]. In addition, the temperature positions are also opposite to those of the RDL in Fig. 15(b), which is not consistent with the placement of the four chiplets. The reason for these problems is probably that the temperature variations are significantly moderated due to the coarse trapezoidal grid cells of the heat spreader and the heat sink. In contrast to these phenomena, the temperature variation in Fig. 17(a) is obvious, and the temperature positions are consistent with those in Fig. 15(a). This is because the grid cells of the peripheral regions are consistent with those of the stacked layers in the proposed thermal model. Specifically, the grid cells of the heat spreader are set to 128 × 128, and the grid cells of the heat sink are set to 256 × 256. This makes the simulation results more accurate, which has been confirmed by the validation against FEM in [19]. Nevertheless, it is at the cost of longer runtime. With a computer with Intel Core i5-6200U CPU (2.3 GHz) and 8 GB RAM, HotSpot takes less than 9 s, while the runtime of the proposed thermal model is 84 s.

VII. Implementation

For one CHI system, the previous isotropic thermal model and the current anisotropic thermal model are utilized to perform package-scale thermal simulations, as demonstrated in this section. Simulation parameters and boundary conditions are presented in Section VII-A. Section VII-B makes a contrast and analysis of the package-scale temperature fields of the two thermal models.

TABLE II
Dimensions and Thermal Conductivities

NAME	SIZES (mm)	CONDUCTIVITY ($\frac{w}{mK}$)
Solder Bumps	50 × 50 × 0.3	2.8(n), 1.6(p)
Organic Substrate	50 × 50 × 1.5	0.33
C4 Bumps	30 × 30 × 0.02	NA
TSVs	30 × 30 × 0.1	NA
RDL	30 × 30 × 0.005	NA
Microbumps	9 × 9 × 0.02	NA
Chiplets	9 × 9 × 0.15	130
TIM	9 × 9 × 0.1	1.6
Base	50 × 50 × 1	237
Fins	2 × 50 × 4, space = 2.7	237

TABLE III
Design Parameters of TSVs, Microbumps, C4 Bumps, and RDL

NAME	SIZES (μm)	SIZE, PITCH (μm)
TSVs	$r_m = 25$, $t_i = 1$	300 × 300, 100
C4 Bumps	$r_m = 25$	300 × 300, 100
Microbumps	$r_m = 25$	150 × 150, 60
RDL1	l = 8000, w = 20	p = 28
RDL2	l = 8000, w = 10	p =26
RDL3	l = 8000, w = 10	p =18
RDL4	l = 8000, w = 10	p= 14

A. Simulation Parameters and Boundary Conditions

Fig. 4 presents the typical architecture of CHI systems. In this section, a CHI system is parameterized as follows: the dimensions and thermal conductivities of solder bumps, organic substrate, four chiplets, TIM, and the heat sink are displayed in Table II. n and p refer to normal and planar directions. All the parameters in Table II come from or are set according to [17], [19], [25], [27], and [28].

As for TSVs, bumps, and RDL, the design parameters should be defined to extract their thermal conductivities. Therefore, the design parameters of microbumps, RDL, TSVs, and bumps are shown in Table III, where r_m, t_i, l, and w refer to the radius of metal cylinders or spheres, the thickness of insulators in TSVs, and the length and width of mental wires, respectively. The design parameters of TSV and bumps are set according to [17].

As for RDL, the design parameters are extracted from a GDSII layout file, as shown in Fig. 18. In this GDSII layout, all the metal wires are parallel underneath chiplets 1, 2, 3, and 4 for the sake of model simplification. Hence, the RDLs 1, 2, 3, and 4 refer to metal wires under corresponding chiplets.

The thermal conductivity of copper in TSVs, microbumps, C4 bumps, and RDL is set to 401 $w/(mK)$. The thermal conductivity of silicon dioxide in TSVs is 1.5 $w/(mK)$. The substrates of TSVs, microbumps, and RDL are silicon, underfill, and silicon dioxide. They are set to 130, 0.5, and

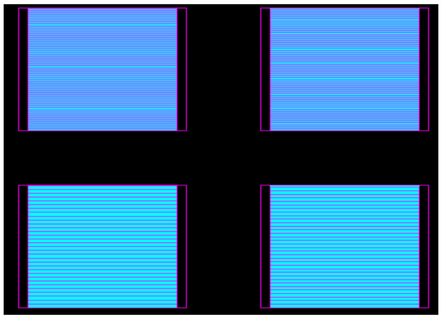

Fig. 18. GDSII layout of metal wires in the RDL.

TABLE IV

ISOTROPIC THERMAL CONDUCTIVITIES OF TSVS, BUMPS, AND RDL

NAME	CONDUCTIVITY ($\frac{w}{mK}$)
TSVs	177.06 (n), 87.11 (p)
C4 Bumps	66.73 (n), 0.73 (p)
Microbumps	184.47 (n), 2.07 (p)
RDL	77.17 (n), 2 (p)

1.5 $w/(mK)$, respectively. The aforementioned thermal conductivities are intrinsic physical parameters of different materials and have been commonly used in thermal simulations [5], [17], [19], [20], [21], [26].

In addition, diabatic surfaces exposed to the ambient air (25 °C) are highlighted in red lines in Fig. 4 with the heat transfer coefficient $h = 70\ W/(m^2K)$. The power consumption of each chiplet is set to 10 W. These thermal parameters are set according to [19].

For this parameterized CHI system, both isotropic and anisotropic thermal simulations have been performed as a demonstration. The differences between the two types of thermal simulations are the ways in which the thermal conductivities of TSVs, RDL, and bumps are extracted. In the isotropic thermal model, the thermal conductivities of TSVs, bumps, and RDL are obtained as follows.

TSV, bump, and RDL layers are regarded as homogeneous blocks with isotropic thermal conductivities. With the equivalent thermal conductivity algorithm in [17], the isotropic thermal conductivities of TSV and bump blocks are extracted according to their design parameters in Table III and the intrinsic thermal conductivities of materials. The thermal conductivity of the RDL block is derived using (9) and (12) with just one grid cell, which means isotropic calculation. Finally, the isotropic thermal conductivities of TSVs, bumps, and RDL are displayed in Table IV.

In the anisotropic thermal simulation, the anisotropic thermal conductivities are extracted with the proposed anisotropic methodologies and according to the design parameters which are displayed in Table III. Therefore, the anisotropic distributions of thermal conductivity are illustrated in Figs. 19 and 20.

B. Simulation Results and Discussion

The two thermal profiles of the RDL are shown in Fig. 21 for comparison. Apparently, the present anisotropic model obtains a larger temperature range than the isotropic model. Moreover, in the high-temperature areas, regions in red differ slightly. In the peripheral regions, the temperatures in (a) are lower than 95 °C while those in (b) are higher than 100 °C, which can be a huge variation. Correspondingly, the global package-scale steady-state temperature field by the proposed anisotropic thermal model is shown in Fig. 22.

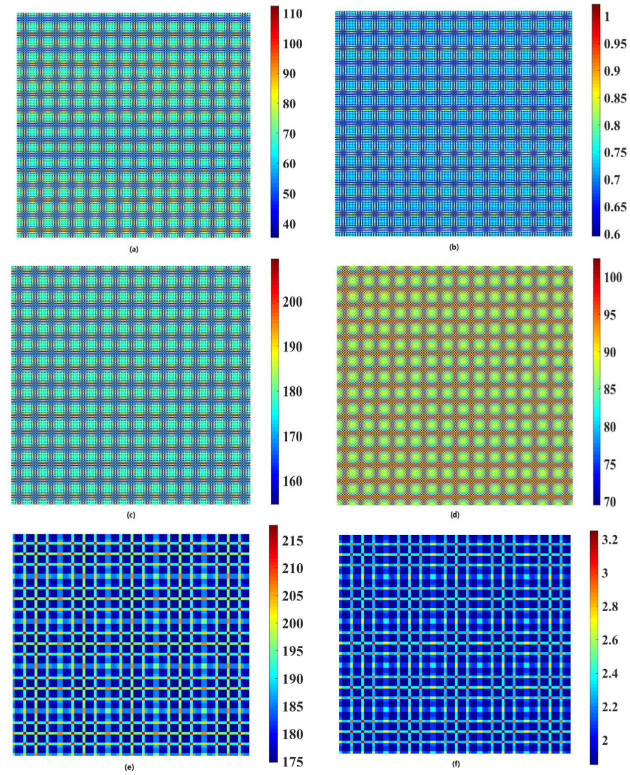

Fig. 19. Anisotropic distributions of (a), (c), and (e) normal and (b), (d), and (f) planar thermal conductivity of (a) and (b) C4 bumps, (c) and (d) TSVs, and (e) and (f) microbumps.

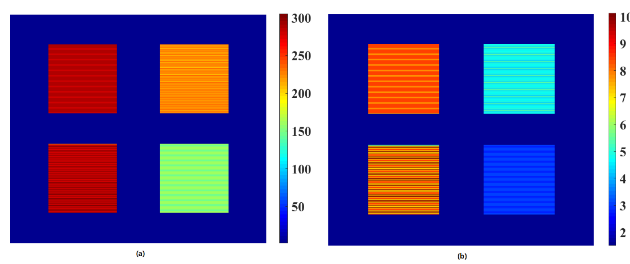

Fig. 20. Anisotropic distribution of (a) normal and (b) planar thermal conductivities of RDL.

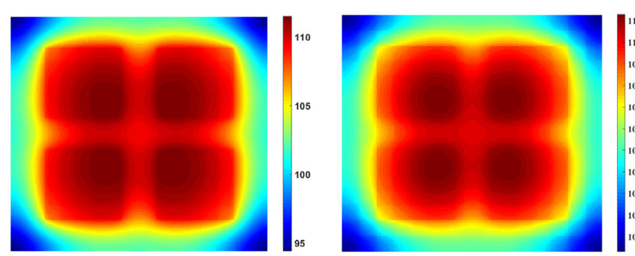

Fig. 21. Thermal profiles of RDL derived by (a) anisotropic and (b) isotropic thermal models.

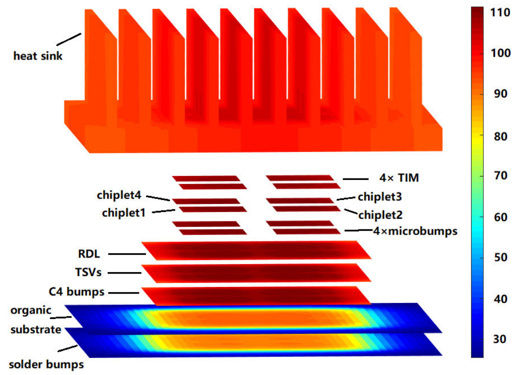

Fig. 22. Global package-scale steady-state temperature field by the anisotropic thermal model.

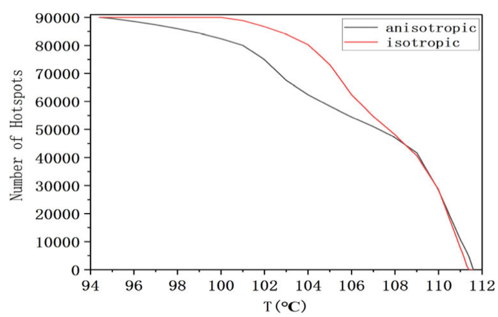

Fig. 23. Number of hotspots detected by the anisotropic and isotropic thermal models under different temperature thresholds.

In the present simulation, hotspots can be defined as temperatures higher than a threshold. Based on this definition, the detected hotspot numbers are different, as shown in Fig. 23. If a temperature T, meaning the abscissa axis, is considered as the temperature threshold, the ordinate axis represents the number of hotspots with temperatures higher than T. It can be noted that the number of grid cells in RDL is 90 000. As illustrated in Fig. 23, if the temperature threshold is set to a value higher than 110 °C, nearly three thousand (more than 3.33% of the grid cells) extra overheating hotspots can be recognized by the current anisotropic model. However, these hotspots escape from detection if we use the isotropic thermal model. On the other hand, if the temperature threshold is set to a value lower than 108 °C, over ten thousand "innocent" grid cells (more than 11.1% of the grid cells) can be diagnosed by the isotropic thermal model as hotspots by mistake since they are actually not overheating. This would impose more pressure on thermal optimization and lead to conservative decisions and extra efforts, which seem plausible but actually redundant.

The result differences between the two thermal models may be explained as follows: in the isotropic thermal model, the uniform equivalent thermal conductivity presupposes the homogenization of TSVs, bumps, and RDL. However, the assumption prevents isotropic thermal models from scrutinizing the local microscopic design structures and the impacts of the local thermal conductivities. As a result, homogenization eliminates the gradient of the thermal conductivity in these complicated components. In other words, the high local thermal conductivity is lower to the uniform equivalent thermal conductivity, and the low local thermal conductivity is heightened to the uniform equivalent thermal conductivity. Consequently, the upper and lower limits of the temperature field are both mitigated. Specifically, compared with the present anisotropic thermal model, the temperatures of the peripheral regions (the lower limit of the temperature field) are heightened over 5 °C, as shown in Fig. 21(b). In addition, the number of hotspots higher than 111 °C (the upper limit of the temperature field) decreases by nearly 3000 (more than 3.33% of the number of grid cells), as shown in Fig. 23. Therefore, the nature of homogenization should be responsible for the deficiency of isotropic thermal models.

TABLE V
JUNCTION TEMPERATURE DIFFERENCES BETWEEN CHIPLETS IN THE ANISOTROPIC THERMAL MODEL

NUMBER	T_j(°C)	NUMBER OF HOTSPOTS
1	111.5	1164
2	111.5	1497
3	111.5	1276
4	111.5	1162

In Fig. 21(a), although the junction temperatures of four chiplets are the same as 111.5 °C, hotspot numbers are not the same as shown in Table V due to the differences in widths and pitches of the metal wires in the RDL. Each chiplet has 21 675 grid cells in total. It implies that a larger width or lower pitch in the RDL design is conducive to heat dissipation in the CHI systems. The reason is that larger width or lower pitch increases the percentage of the metal, which is greatly capable of heat conduction, and thus improves the heat dissipation environments.

VIII. Conclusion

In this work, a multiscale anisotropic thermal model for the CHI systems is developed to predict the package-scale steady-state temperature profiles. It enables the extraction of the design features from GDSII layout files and the calculation of the local material compositions and thermal conductivities. In the steady-state heat equation, thermal conductance matrix $\boldsymbol{G}$ can be optimized by the anisotropic distributions of the thermal conductivity. While considering the anisotropy of bumps, TSVs, and RDL, other components can still be regarded as isotropic blocks to obtain global package-scale steady-state temperature fields of the CHI systems.

As evaluated in Sections VI and VII, the proposed anisotropic thermal model can obtain more accurate temperature maps and discover more reasonable hotspots than the previous isotropic thermal model. Without the anisotropic thermal model, the obscure hotspots that survive the detection of the isotropic thermal model may lead to unexpected thermally related reliability problems, even failure in products. Meanwhile, lots of "innocent" grids can avoid being recognized as hotspots by mistake, cutting down the cost of thermal management and optimization in the design flow of the CHI systems.

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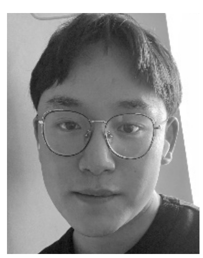

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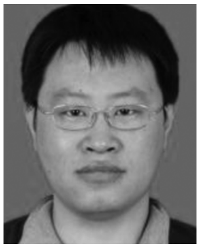

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